MoonWalker Weekly Meeting

1/6 - 1/13 Summary

Elijah, Rowel, Martin

Progress Update

- Martin
  - Working out the small kinks during collision demos
  - Reading material related to reinforcement learning as well as Haolin’s paper
  - Working with Rowel’s omniverse setup
- Elijah
  - Implemented Collider and Velocity parameter extraction
  - Generalized scripts to allow for better configuration setup
- Rowel
  - Set up Omniverse Environment in Digital Twin Server
  - Designating Host in API
  - Updating HSML-Form-Website to allow users to pick Host

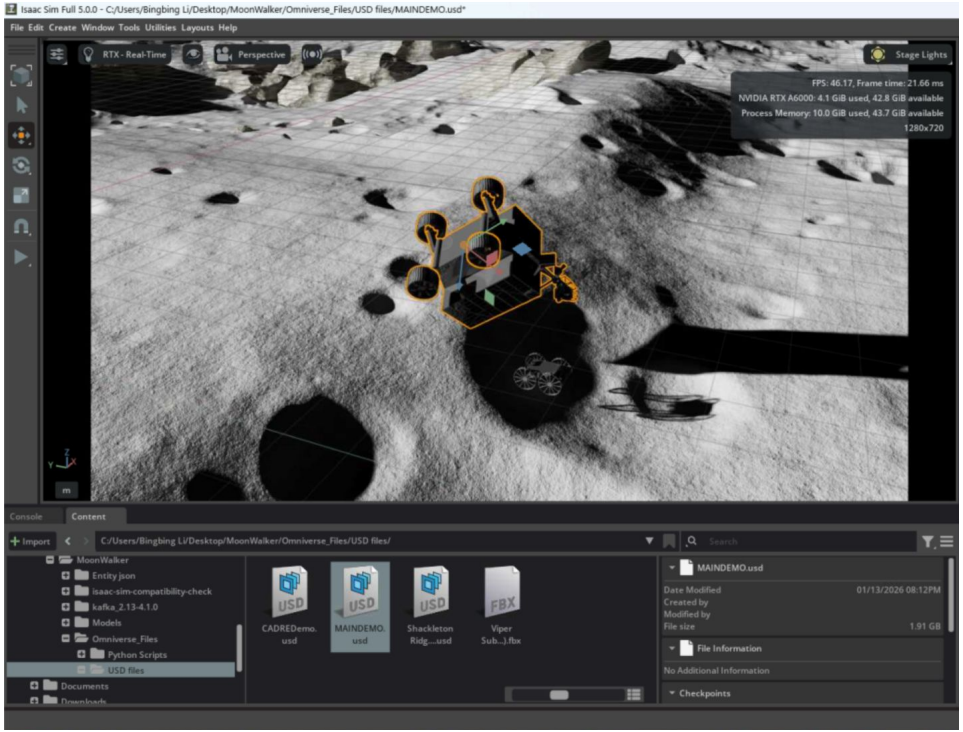
Working out the kinks: Gimbal lock

```
        ]
    },
    "center_of_mass": [
        -Infinity,
        -Infinity,
        -Infinity
    ],
    "linear_velocity": [
        -0.009112503379583359,
        0.7772644758224487,
        0.27406296133995056
    ],
    "angular_velocity": [
        5.881232738494873,
        24.728801727294922,
        -14.727397918701172
    ],
    "colliders": [
        {
            "prim_path": "/World/Omni_Cadre/CADRE_Demo/Chassis",
            "type_name": "Mesh",
            "mesh_approximation": "convexHull"
        }
    ],
    "last_collision": {
        "actor0": "/World/Omni_Cadre/CADRE_Demo/FR",
        "actor1": "/World/Unity_Viper/Xform/Chassis",
        "collider0": "/World/Omni_Cadre/CADRE_Demo/FR",
        "collider1": "/World/Unity_Viper/Xform/Chassis",
        "timestamp": "2026-01-14T00:28:55.710466"
    },
    "hosting": false,
    "modifiedDate": "2026-01-14T00:31:09.900519"
}
```

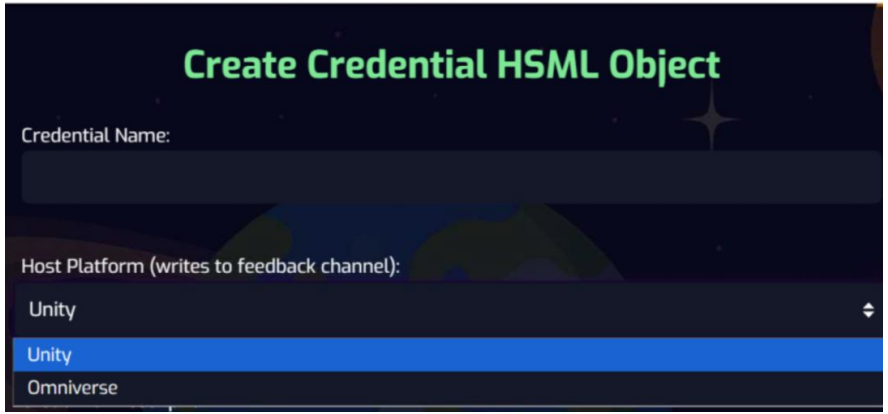

Updated JSON file from Omniverse with velocities, collider and collision information

```
#============= USER CONFIG =============
CONFIG = {
    # USD prim paths
    "prim_paths": {
        # Chassis Prim of Rover script is attached to
        "self_chassis": "/World/Omni_Cadre/CADRE_Demo/Chassis",

        # Chassis Prim of the other rover that is in the world
        "other_rover_chassis": "/World/Unity_Viper/Xform/Chassis",

        # Wheel Joint Prims
        "wheel_joints": [
            "/World/Omni_Cadre/CADRE_Demo/FR/RevoluteJoint",
            "/World/Omni_Cadre/CADRE_Demo/FL/RevoluteJoint",
            "/World/Omni_Cadre/CADRE_Demo/BR/RevoluteJoint",
            "/World/Omni_Cadre/CADRE_Demo/BL/RevoluteJoint",
        ],
    },

    # Joint Drive Configuration
    "drive": {
        "velocity_attr": "drive:angular:physics:targetVelocity",
        "default_velocity": 400.0,      # used in dynamic mode
        "stop_velocity": 0.0,           # used in kinematic mode
    },

    # Producer/Consider JSON
    "io": {
        # Producer output JSON
        # use_tempdir=True: file is written to OS temp dir with filename below
        # use_tempdir=False: set output_path to an absolute path
        "producer_output": {
            "use_tempdir": True,
            "filename": "producerOmniUpdate.json",
            "output_path": "",          # only used if use_tempdir=False
        },
```

User Configuration dictionary for ease of use



Lunar Environment and Rovers in DT Server



Allows you to pick host when creating a credential

```
moonwalker@DESKTOP-TOD84J7:~/kafka_2.13-4.1.0$ bin/kafka-acls.sh --bootstrap-server 172.26.30.154:9092 --list
Error while executing ACL command: org.apache.kafka.common.errors.SecurityDisabledException: No Authorizer is configured
 on the broker
```

Kafka is not using ACL only MySQL authentication

Action Plan

- Rowel
  - Work with Martin Testing out the update Physics parameters in Omniverse
    - Work on Cadre producing script for Viper (when Omniverse is Host)
  - More Research on EC2 and move toward remote access testing
  - Update Kafka to implement ACL (Optional)
- Elijah
  - Finish Generalization Script
  - Testing Updated Scripts
- Martin
  - Assist Rowel with setting up simulations and testing
  - Clean up demo testing (i.e make it look better)
  - Continue reading Haolin’s paper and learning reinforcement learning
  - Check out Cosmos